INCH-POUND

MIL-C-70996 (AR)
9 May 1991

MILITARY SPECIFICATION

CARTRIDGE, HE, 81MM, M821E1/M889E1 W/FUZE M734/M935 LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for loading, assembling and packing for two types of mortar cartridges designated as Cartridge, 81MM, M821E1 and Cartridge, 81MM, M889E1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1315

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STANDARDS

MILITARY

MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic and Ultrasonic)
MIL-STD-453	- Inspection, Radiographic
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of Federal and Military Specifications, Standards and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

12630535	- Cartridge, 81mm: M821E1, with Fuze, Multi-Option: M734
12630600	- Packing and Marking for Cntnr, PA 149 Ammunition Fiber for Cartridge, 81MM, M821E1, M889E1 and M879
12630601	- Parking and Marking for Box, Wirebound for Cartridge 81MM, M879, M821E1 and M889E1
12630672	- Cartridge, 81MM: HE, M889E1 with Fuze, PD: M935

(Copies of drawings and other publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable specifications, drawings and this specification.

3.1.1 Clean explosive scrap. Clean scrap (see 6.8) produced in regular operations may be remelted and reused provided that it complies with the requirements, except for form, for the grade of high explosive specified. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used.

3.2 Cartridge. The cartridge shall comply with all requirements specified on drawing 12630535/12630672 and associated drawings and with all requirements specified in applicable standards and specifications.

3.3 Surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The maximum (max.) dimension of any visible cavity in the faced (bottom) surface of the fuze well shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. (Cavities having a max. dimension of 1/16 inch or less shall be disregarded in this computation).

3.4 Foreign matter in projectile cavity. There shall be no foreign matter in the projectile cavity prior to high explosive pouring. Solid explosive shall not be present.

3.5 High explosive on threads. Threads shall be free of high explosive.

3.6 High explosive on projectile body. The exterior surface of the projectile body shall be free of high explosive.

3.7 Cavitation in high explosive charge (see 4.5.3).

3.7.1 General. Cavities having a maximum projected dimension of 3/32 inch or less shall be disregarded provided that they are separated by 1/2 inch or more of solid explosive.

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3.7.2 Maximum projected areas and dimensions of cavities. The aggregate of the projected area of all cavities shall not exceed 1 square inch and no individual cavity shall exceed 1/2 square inch. Cavities located in the area of the charge which extends from the base of the fuze well to 2.5 inches below the base of the fuze well shall not exceed 3/4 inch in any direction and their aggregate projected area shall not exceed 5/8 square inch.

3.7.3 Piping cavity. A piping cavity on the longitudinal axis of the projectile shall be permitted provided that its projected length does not exceed 1 1/2 inches and its width does not exceed 1/4 inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.7.4 Cracked charges and annular rings. Not more than four transverse cracks shall be permitted in any charge. The cracks shall be no wider than 1/32 inch. Longitudinal cracks shall be disregarded. The presence of annular rings with a projected width of 1/16 inch max. and extended from the shell wall 1/4 inch max. into the charge shall not be cause for rejection.

3.7.5 Base separation. The base separation from the inside of the rear of the shell to the base of the explosive shall be not more than 1/32 inch.

3.8 Ballistic performance (proving ground). The cartridges shall comply with the following requirements when tested as specified in 4.5.1.

3.8.1 Misfire. The cartridges shall not fail to fire from the weapon.

3.8.2 Duds. The cartridges shall function on impact.

3.8.3 Short rounds. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range when calculated without the short round(s).

3.8.4 Premature burst. There shall be no premature burst in the gun bore or in the first 100 meters of flight.

3.8.5 Early function. There shall be no early function after the first 100 meters of flight and prior to impact functioning.

3.8.6 Metal parts separation. There shall be no metal parts separation in the gun bore or in-flight.

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3.8.7 Range precision. The horizontal range standard deviation shall be no greater than 22 meters when fired at charge zero and no greater than 83 meters when fired at charge four.

3.9 Workmanship. Parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer in accordance with the provisions of 4.3.2. The first article sample shall consist of the items in the quantities indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I.

4.3.3 Rejection. See MIL-A-48078 and the first article ballistic test shall be rejected if:

a. One or more misfires occur.

b. Five or more duds occur.

c. One or more short rounds occur.

d. One or more premature burst occur.

e. One or more early functions occur.

f. One or more metal parts separation occur.

g. Horizontal range standard deviation is exceeded ($70^{o}F$ conditioned samples only).

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TABLE I First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70996 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, 81MM, HE, M821E1/M889E1	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Body Loading Assembly (dwg. 12630553)			
	Examination for defects	25	3.2	4.4.2.2
	Cartridge (dwg. 12630535/12630672)			
	Examination for defects	25	3.2	4.4.2.4
	Proving Ground	144	3.8	4.4.3.1/4.5.1

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots of cartridges shall contain:

a. Projectile metal parts assemblies of one lot interfix from one manufacturer.

b. Composition B of one lot interfix from one manufacturer.

c. Fin assemblies of one lot interfix from one manufacturer.

d. Ignition cartridges of one lot interfix from one manufacturer.

e. Propelling charges of one lot interfix from one manufacturer.

f. Obturating rings of one lot interfix from one manufacturer.

g. Fuzes of one lot interfix from one manufacturer.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Where specified herein, attributes sampling inspection shall be conducted in accordance with TABLE II below, using the inspection levels cited in the Classification of Characteristics paragraphs:

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TABLE II. Sampling

LOT SIZE	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative inspection provisions. Alternative inspection procedures, methods or equipment such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval (see 6.10). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.1	TITLE Projectile, HE, M821E1 or M889E1 Body Assembly	SHEET 1 OF 1	DRAWING NUMBER 12630554 / NEXT HIGHER ASSEMBLY 12630553
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Foreign object in charge cavity	100%	3.4	Visual
Major	None defined			
Minor				
201	Interior protective coating with bare spot	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.9	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body Loading Assembly	1 OF 1	12630553
			NEXT HIGHER ASSEMBLY: 12630535/12630672

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	High explosive on projectile body	100%	3.6	Gage
Major				
101	Depth of fuze well	Level III	3.2	Gage
102	Diameter of fuze well	Level III	3.2	Gage
Minor				
201	Surface cavitation excessive	Level V	3.3	Visual
202	High explosive on threads	Level V	3.5	Visual
203	Evidence of poor workmanship	Level V	3.9	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70996 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Charge, Propelling	1 OF 1	9381510
			NEXT HIGHER ASSEMBLY 12630535/12630672

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Container or container seal damaged to the extent that propellant will escape	100%	3.2	Visual/Manual (see 4.5.4)
Major				
101	Container or container seal damaged but not to the extent that propellant will escape	100%	3.2	Visual/Manual (see 4.5.4)
102	Gross weight of propelling charge, min.	100%	3.2	Gage
Minor	None defined			

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cartridge	1 OF 2	12630535/12630672
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Ignition cartridge not properly assembled or fully seated	100%	3.2	Gage
2	Propelling charge missing or damaged to the extent that propellant will escape	100%	3.2	Visual/Manual (see 4.5 4)
3	Obturating ring missing 1/	100%	3.2	Gage
4	Pull wire missing or not fully seated 2/	100%	3.2	Gage
Special				
A	Warning statement missing or illegible	100%	3.2	Visual
B	Fin blade distorted	100%	3.2	Gage

NOTES: 1/ This inspection may be performed at a prior station when deemed necessary.
2/ This inspection applies only to the M889E1 Cartridge with the M935 fuze.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cartridge (Continued)	2 OF 2	12630535/12630672
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
101	Obturating ring not flush or below bourrelet diameter 1/	100%	3.2	Gage
102	Outside diameter, max.	100%	3.2	Gage
103	Cartridge assembly fail to fully enter profile and alignment gage	100%	3.2	Gage
104	Obturating ring damaged	Level III	3.2	Visual
105	Fin assembly torque	Level III	3.2	Test
106	Fuze torque	Level III	3.2	Test
107	Ignition cartridge torque	Level III	3.2	Test
108	Fin assembly not fully seated	Level III	3.2	Visual
109	Fuze not fully seated	Level III	3.2	Visual
110	Propelling charge damaged but not to the extent that propellant will escape	100%	3.2	Visual/Manual (see 4.5.4)
Minor				
201	Paint damage area in excess of 1/4 square inch	Level III	3.2	Visual
202	Marking missing, misleading or unidentifiable	Level V	3.2	Visual
203	Pettman cement application improper	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.9	Visual
	Proving Ground Test			
	Ballistic Acceptance	48	3.9	4.4.3.1/4.5.1

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Packing & Marking for Cntnr, PA 149, Ammunition Fiber for Cartridge, 81MM, M821E1, M889E1 & M879 (Unsealed)	1 OF 1	12630600
			NEXT HIGHER ASSEMBLY 12630601

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Propelling charge support assembly missing or improperly positioned	100%	5.1	Visual
Major				
101	Fuze support missing	Level III	5.1	Visual
102	Cushion missing	Level III	5.1	Visual
103	Fiber container parts not secure	Level III	5.1	Manual
Minor				
201	Evidence of poor workmanship	Level V	5.1	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70996 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Packing & Marking for Cntnr, PA 149, Ammunition Fiber for Cartridge, 81MM, M821E1, M889E1 & M879 (Sealed)	1 OF 1	12630600
			NEXT HIGHER ASSEMBLY 12630601

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged	Level III	5.1	Visual
102	Sealing strip incomplete, badly wrinked or missing	Level III	5.1	Visual
103	Contents loose	Level III	5.1	Manual
Minor				
201	Marking missing, misleading or unidentifiable	Level V	5.1	Visual
202	Evidence of poor workmanship	Level V	5.1	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.7	TITLE Packing & Marking for Box, Wirebound for Cartridge, 81MM, M879, M821E1, and M889E1	SHEET 1 OF 1	DRAWING NUMBER 12630600
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Containers loose	Level III	5.1	Visual
102	Containers improperly oriented	Level III	5.1	Visual
Minor				
201	Firing table not present in every four boxes	Level V	5.1	Visual
202	Plastic sheet not present in every six boxes	Level V	5.1	Visual
203	Marking missing, misleading or unidentifiable	Level V	5.1	Visual
204	Evidence of poor workmanship	Level V	5.1	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Ballistic testing. The cartridges shall be observed for the following characteristics:

a. Misfires, Major defect (see 3.8.1).

b. Duds, Major defect (see 3.8.2).

c. Short rounds, Critical defect (see 3.8.3).

d. Premature burst, Critical defect (see 3.8.4).

e. Early function, Major defect (3.8.5).

f. Metal parts separation, Special defect (see 3.8.6).

g. Range precision, Major defect (see 3.8.6).

Testing shall be conducted as specified in 4.5.1.

4.4.3.1.1 First article. A sample of 144 cartridges shall be used for this test. Accept/reject criteria shall be in accordance with 4.3.3.

4.4.3.1.2 Production. A sample of 48 cartridges shall be randomly selected from each lot for this test. The lot shall be rejected if:

a. One or more misfires occur.

b. Three or more duds occur.

c. One or more short rounds occur.

d. One or more prematures occur.

e. One or more early functions occur.

f. One or more metal parts separation occur.

g. Horizontal range standard deviation is exceeded.

4.4.3.2 Strength of heat weld. One obturating ring sample, minimum, after every one hundred (100) projectiles, shall be assembled to a fixture, introduced to the production line, heat sealed and tested as specified in 4.5.2. Failure of the sample to meet the requirement shall be cause for those one hundred (100) preceeding projectiles represented by the sample to be rebanded and retested.

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4.4.3.3 Cavitation (see 3.7) major defect. The sampling plan and procedure for this test shall be in accordance with MIL-STD-1235 using CSP-2 plan having the following parameters; i value shall be 144, f value shall be 1/24, and the S value shall be 1800. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.5.3.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.5 herein.

4.5 Methods of inspection.

4.5.1 Proving ground test. The test shall be performed at a Government proving ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The cartridges for first article testing shall be separated into three equal groups; one group conditioned to 145 $\pm$ 2.5°F for 24 hours, one group conditioned to 70 $\pm$ 2.5°F for 24 hours and one group conditioned to -50 $\pm$ 2.5°F for 24 hours. Half of each temperature group will be tested at charge zero and half will be tested at charge four.

The cartridges for production lot acceptance shall be conditioned at 70°F $\pm$ 2.5°F. The sample shall be separated into two equal groups; one group tested at charge zero and one group tested at charge four.

The cartridge shall be fired from a M252 Mortar at an elevation of forty-five degrees from a hard mount (a solid base test fixture) within two minutes after removal from the conditioning chamber. All fuzes shall be set for impact functioning (M734 - "IMP" and M935 - "SQ"). The individual velocities, pressures, deflections, and ranges shall be recorded on the firing record.

Observation shall be made for misfires, earlies and duds, and shall be recorded. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence

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and attached to the firing record. The velocity and range standard deviationsand arithmetic averages shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded.

Hawk radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of these rounds, and prints of the preceding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded.

A smear camera shall be used to photograph all test samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all the other rounds in that particular charge zone, shall be attached to the firing record. Film from acceptable lots may be discarded.

4.5.1.1 Test validity. If for any reason the proving ground personnel considers that the test conditions have detrimentally affected the test result, the test result shall be declared invalid and a new test shall be performed on additional sample cartridges.

4.5.2 Strength of heat weld. The sample shall be tested for integrity by applying the required load at the heat weld radially and in a direction away from the axis (center) of the ring.

4.5.3 Determination of cavities. The sample cartridges (HE filled) shall be subjected to radiographic examination. The radiographic beam shall form a 90° angle with the longitudinal axis of the projectile. Examinations of the radiograph shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of the radiographic technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.5.3.2. In the event an original radiograph leaves doubt as to acceptablitity, the cartridge shall be re-radiographed at an angle 90° from that of the original exposure and in the same plane.

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4.5.3.1 Sensitivity check of radiographic technique. The sensitivity of radiographic technique shall be determined by the means of a penetrameter fabricated as specified herein. Location of the penetrameter hole shall be 6 3/4 inches measured from base shoulder of projectile body towards fuze nose threads. The projectile, with attached penetrameter, shall be positioned between the radiographic source and the radiographic film with the penetrameter facing the radiographic source. When radiographing a group of projectiles on a rotating ring, the penetrameter, taped to a projectile, may be used to represent the group when checking sensitivity. The sensitivity of radiographic technique shall be considered satisfactory when the image of the penetrameter, including the holes, is sharply defined on the radiograph. The penetrameters shall be made of steel having an approximate specific gravity of 7.8. Penetrameter thickness shall be .020 plus or minus (+) .010 inches. Each penetrameter shall have a circular hole with a diameter twice that of the penetrameter thickness. For further details on construction of the penetrameters, see MIL-STD-453. Frequency of sensitivity check shall be no less than one in each fifty (50) radiographs.

4.5.3.2 Check of photographic density of flim. The photographic density of the film shall be determined on the film in the region of the maximum equivalent thickness to be radiographed. Film density shall be checked each hour of operation, or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. To judge densities, it is recommended that standards be used. Comparison of the film with standards may be made visually; hovever, the use of a densitometer is recommended.

4.5.3.3 Personnel qualification and certification. Personnel performing the radiographic examination in accaordance with paragraph 4.5.3 shall meet the qualification and certification requirements defined in MIL-STD-410 as follows:

4.5.3.3.1 Set up and calibration. Level II.

4.5.3.3.2 Operator. Level I.

4.5.3.3.3 Documentation. The contractor will prepare and submit for approval, detailed procedures for qualification and certification of personnel performing the non-destructive testing procedures required herein in accordance with the approved Data Item Description specified on the Contract Data Requirements List in the contract (see 6.3).

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4.5.4 Damaged propelling charge. To ascertain that a damaged propelling charge is to be classed as critical or major, the following procedure shall be used:

a. Place the damaged charge on a M24 fin, oriented such that the propellant would be resting against the damaged surface.

b. The fin-charge shall be hand-shaken vigorously three times.

c. Examination shall be made for propellant spillage - if any propellant is observed, the damaged propelling charge shall be classed as a critical defect; if no propellant is observed, the damaged propelling charge shall be classed as a major defect.

5. PACKAGING

5.1 Unit packaging.

5.1.1 Level A. Packaging for cartridge shall be in accordance with the requirements of drawing 12630600.

5.1.2 Level B. Same as level A.

5.1.3 Level C. Same as level A

5.2 Packing.

5.2.1 Level A. Packing for cartridge shall be in accordance with the requirements of drawing 12630601.

5.2.2 Level B. Same as level A.

5.2.3 Level C. Same as level A.

5.3 Marking.

5.3.1 Level A. Unit packaging for cartridge shall be marked in accordance with the requirements of drawing 12630600. Packing shall be marked in accordance with the requirements of drawing 12630601.

5.3.2 Level B. Same as level A.

5.3.3 Level C. Same as level A.

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6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for assembly to M821E1/M889E1 cartridges for use with the 81MM mortar.

6.2 Acquisition requirements.

a. See MIL-A-48078.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Data requirements. The following Data Item Descriptions (DIDs) must be listed as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID NUMBER	DID Title
4.5.3.3.3	DI-S-3628	Nondestructive Inspection Plan

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited in the DD Form 1423.

(Copies of data item descriptions required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of

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items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Submission of designs for approval. See paragraph 6.2.3 of MIL-A-48078 (AR). Submit equipment designs, as required, to Commander, AMCCOM, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List in the contract.

6.6 Drawings. Drawings listed is Section 2 of this specification under the heading U.S. Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following:

Commander
U.S. Army Armament Research, Development and
Engineering Center
ATTN: AMSMC-QAT-B (D)
SMCAR-FSA-IM
Picatinny Arsenal, NJ 07806-5000

6.8 Clean scrap. Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation. Scrap from floor sweepings, ventilation, and suction apparatus (other than that of drilling operations) is not defined as being clean scrap and shall not be used.

6.9 Loading of high explosive. Solid explosive shall not be added to the melted explosive or to the projectile body during pouring or cooling.

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6.10 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.11 Subject term (key word) listing.

Ammunition, mortar
Ballistics
Explosive, high
Radiography

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AD04)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70996 (AR)	9 May 1991

3. DOCUMENT TITLE: CARTRIDGE, HE, 81MM, M821E1/M889E1 W/FUZE M734/M935 LAP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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